Michigan Tech

Presenting Computational Mathematics on a Poster

Show the result and how it was computed

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Math Sciences

MA4535 — Nonlinear Dynamics & Chaos | Lecture 9 | September 23, 2026

Show the computational argument

- Lead with one question, one result, and the strongest supporting figure.
- State the model, parameter values, and initial conditions.
- Name the numerical method, time interval, and solver settings.
- Show a check: finer resolution, tighter tolerance, or independent data.
- Say what the computation establishes and where it falls short.

Make the main figure reproducible

- Use several figures. Caption each with its result; label axes, curves, and units.
- Show brief pseudocode; link the full code and data.

Example pseudocode for a phase portrait

$p, \boldsymbol{y}_0 \leftarrow$ parameters, initial state
$Y \leftarrow \textsc{SolveODE}(f, p, \boldsymbol{y}_0, [0, T], \varepsilon)$
$Y \leftarrow \textsc{DiscardTransient}(Y, t_0)$
$\textsc{PlotAndSave}(Y_x, Y_y)$

- Check that labels and the code remain readable at poster size.

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Local Flow Geometry

Local geometry of the Rössler flow

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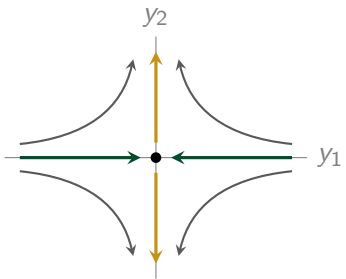
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Equilibrium points, again

An **Equilibrium Point** $\boldsymbol{q}^*$ of $\boldsymbol{y}' = \boldsymbol{f}(\boldsymbol{y})$ satisfies $\boldsymbol{f}(\boldsymbol{q}^*) = \boldsymbol{0}$ and $\boldsymbol{y}(t) = \boldsymbol{q}^*$.

A linear saddle is a simple example.

$$\boldsymbol{y}' = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} \boldsymbol{y}.$$



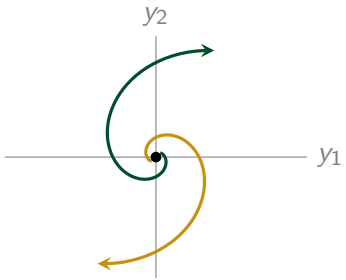
- $(0,0)$ is an equilibrium.
- $(y_1, 0) \xrightarrow[t \to +\infty]{} (0,0)$.
- $(0, y_2) \xrightarrow[t \to -\infty]{} (0,0)$.

Unstable spiral

An **Equilibrium Point** $\boldsymbol{q}^*$ of $\boldsymbol{y}' = \boldsymbol{f}(\boldsymbol{y})$ satisfies $\boldsymbol{f}(\boldsymbol{q}^*) = \boldsymbol{0}$ and $\boldsymbol{y}(t) = \boldsymbol{q}^*$.

A linear spiral is a second simple example.

$$\boldsymbol{y}' = \begin{pmatrix} 1 & 2 \\ -2 & 1 \end{pmatrix} \boldsymbol{y}.$$



- $(0, 0)$ is an equilibrium.
- Nonzero ICs spiral clockwise and outward.
- $\boldsymbol{y}(t) \xrightarrow[t \to -\infty]{} (0, 0)$ for every IC.

Unstable spiral in polar coordinates

For a nonzero IC, write $y_1 = r\cos\theta$, $y_2 = r\sin\theta$, and $r^2 = y_1^2 + y_2^2$. The matrix equation gives $y_1' = y_1 + 2y_2$ and $y_2' = -2y_1 + y_2$.

- Differentiate r^2 and substitute the component equations:

$$rr' = y_1y_1' + y_2y_2' = y_1(y_1 + 2y_2) + y_2(-2y_1 + y_2) = r^2.$$

  The mixed terms cancel, so $r' = r$ and $r(t) = r(0)e^t$.

- Differentiate $y_1 = r\cos\theta$ and $y_2 = r\sin\theta$. For $r > 0$, the angle satisfies

$$r^2\theta' = y_1y_2' - y_2y_1' = y_1(-2y_1 + y_2) - y_2(y_1 + 2y_2) = -2r^2.$$

  Thus $\theta' = -2$ and $\theta(t) = \theta(0) - 2t$.

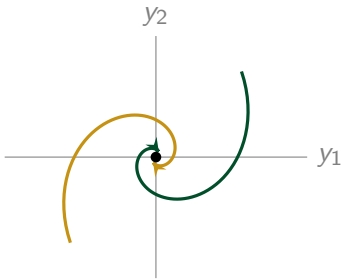
- The radius grows and the angle decreases: trajectories spiral clockwise and outward as t increases.

Stable spiral

An **Equilibrium Point** $\boldsymbol{q}^*$ of $\boldsymbol{y}' = \boldsymbol{f}(\boldsymbol{y})$ satisfies $\boldsymbol{f}(\boldsymbol{q}^*) = \boldsymbol{0}$ and $\boldsymbol{y}(t) = \boldsymbol{q}^*$.

A stable spiral is another simple example.

$$\boldsymbol{y}' = \begin{pmatrix} -1 & 2 \\ -2 & -1 \end{pmatrix} \boldsymbol{y}.$$



- $(0, 0)$ is an equilibrium.
- Nonzero ICs spiral clockwise and inward.
- $\boldsymbol{y}(t) \xrightarrow[t \to +\infty]{} (0, 0)$ for every IC.

Stable spiral in polar coordinates

For a nonzero IC, write $y_1 = r\cos\theta$, $y_2 = r\sin\theta$, and $r^2 = y_1^2 + y_2^2$. The matrix equation gives $y_1' = -y_1 + 2y_2$ and $y_2' = -2y_1 - y_2$.

- Differentiate r^2 and substitute the component equations:

$$rr' = y_1y_1' + y_2y_2' = y_1(-y_1 + 2y_2) + y_2(-2y_1 - y_2) = -r^2.$$

  The mixed terms cancel, so $r' = -r$ and $r(t) = r(0)e^{-t}$.

- Differentiate $y_1 = r\cos\theta$ and $y_2 = r\sin\theta$. For $r > 0$, the angle satisfies

$$r^2\theta' = y_1y_2' - y_2y_1' = y_1(-2y_1 - y_2) - y_2(-y_1 + 2y_2) = -2r^2.$$

  Thus $\theta' = -2$ and $\theta(t) = \theta(0) - 2t$.

- The radius shrinks and the angle decreases: trajectories spiral clockwise and inward as t increases.

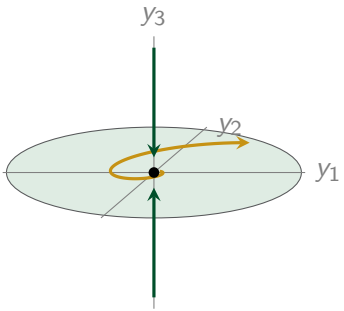
Spiral out, axis in

Add an ingoing axis to the outward spiral.

$$\boldsymbol{y}' = \begin{pmatrix} 1 & 2 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix} \boldsymbol{y}.$$

Let $r = \sqrt{y_1^2 + y_2^2}$ and θ be the polar angle.

- In $y_3 = 0$: $r(t) = r(0)e^t$ and $\theta(t) = \theta(0) - 2t$.
- On the y_3 axis: $y_3(t) = y_3(0)e^{-t}$.
- One stable direction; two unstable directions.



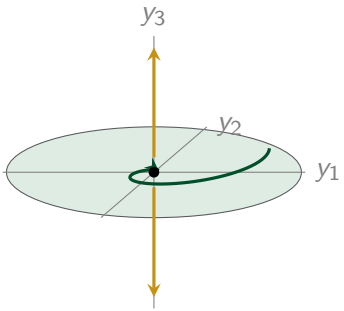
Spiral in, axis out

Add an outgoing axis to the inward spiral.

$$\boldsymbol{y}' = \begin{pmatrix} -1 & 2 & 0 \\ -2 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \boldsymbol{y}.$$

Let $r = \sqrt{y_1^2 + y_2^2}$ and θ be the polar angle.

- In $y_3 = 0$: $r(t) = r(0)e^{-t}$ and $\theta(t) = \theta(0) - 2t$.
- On the y_3 axis: $y_3(t) = y_3(0)e^{t}$.
- Two stable directions; one unstable direction.



Stable and unstable equilibria

Let $\boldsymbol{q}^*$ be an equilibrium of $\boldsymbol{y}' = \boldsymbol{f}(\boldsymbol{y})$, with IC $\boldsymbol{y}(0) = \boldsymbol{y}_0$.

- A **Stable Equilibrium** keeps nearby trajectories nearby: for every $\varepsilon > 0$, some $\delta > 0$ makes

$$\|\boldsymbol{y}_0 - \boldsymbol{q}^*\| < \delta \quad \Longrightarrow \quad \|\boldsymbol{y}(t) - \boldsymbol{q}^*\| < \varepsilon \quad \text{for all } t \geq 0.$$

- An **Unstable Equilibrium** fails this condition: ICs arbitrarily close can leave a fixed neighborhood at a later time.
- Stability means staying close; convergence is a stronger property. The saddle and outward spiral are unstable, while the inward spiral is stable.

Which initial states approach q^*?

Write $\varphi_t(\boldsymbol{y}_0)$ for the trajectory from $\boldsymbol{y}_0$.

- ICs whose trajectories approach $\boldsymbol{q}^*$ as $t \to +\infty$ are in the **Stable Manifold**:

$$W^s(\boldsymbol{q}^*) = \{\boldsymbol{y}_0 : \varphi_t(\boldsymbol{y}_0) \to \boldsymbol{q}^* \text{ as } t \to +\infty\}.$$

- ICs whose trajectories approach $\boldsymbol{q}^*$ as $t \to -\infty$ are in the **Unstable Manifold**:

$$W^u(\boldsymbol{q}^*) = \{\boldsymbol{y}_0 : \varphi_t(\boldsymbol{y}_0) \to \boldsymbol{q}^* \text{ as } t \to -\infty\}.$$

- The equilibrium $\boldsymbol{q}^*$ is in both manifolds.

What is a manifold?

- A **Manifold** looks like $\mathbb{R}^k$ near each point: k local coordinates describe it.
- For $k = 1$ it is locally a curve; for $k = 2$ it is locally a surface.
- A smooth equation $g(\boldsymbol{y}) = 0$ defines a surface (2D manifold) in $\mathbb{R}^3$ where $\nabla g \neq \mathbf{0}$.
  - $g(\boldsymbol{y}) = y_3$ gives a plane.
  - $g(\boldsymbol{y}) = y_1^2 + y_2^2 + y_3^2 - 1$ gives a sphere.
- The equations $y_1^2 + y_2^2 - 1 = 0$ and $y_3 = 0$ give a circle, a 1D manifold.
- Stable and unstable manifolds follow the flow. Near the equilibrium, trajectories remain on them.

Eigenpairs and time direction

Near the equilibrium $\boldsymbol{q}^*$, set $\boldsymbol{u} = \boldsymbol{y} - \boldsymbol{q}^*$. Taylor expansion gives

$$\boldsymbol{u}' = \cancelto{\mathbf{0}}{\boldsymbol{f}(\boldsymbol{q}^*)} + A\boldsymbol{u} + O(\|\boldsymbol{u}\|^2), \qquad A = D\boldsymbol{f}(\boldsymbol{q}^*).$$

Dropping the higher-order term gives $\boldsymbol{u}' = A\boldsymbol{u}$. An eigenpair $(\lambda, \boldsymbol{v})$ of A gives the mode $e^{\lambda t}\boldsymbol{v}$.

- If $\operatorname{Re}\lambda > 0$, the mode grows as $t \to +\infty$; if $\operatorname{Re}\lambda < 0$, it decays.
- As $t \to -\infty$, the behavior reverses:
  $\operatorname{Re}\lambda > 0$ modes decay; $\operatorname{Re}\lambda < 0$ modes grow.

Ergodic Manifold Stable and Unstable Manifold Characterizations

For $A = D\boldsymbol{f}(\boldsymbol{q}^*)$, the linearized equation is $\boldsymbol{u}' = A\boldsymbol{u}$.

- Eigenvalues characterize linear manifolds. E is a mnemonic for eigenspace.
  - E^s is the span of eigenspaces of A with $\operatorname{Re}\lambda < 0$; these modes decay forward in time.
  - E^u is the span of eigenspaces of A with $\operatorname{Re}\lambda > 0$; these modes grow forward in time.
- The equilibrium is **Hyperbolic** when no eigenvalue has zero real part.
- At a hyperbolic equilibrium, the local stable and unstable manifolds are tangent to E^s and E^u, respectively. Each manifold has the dimension of its corresponding subspace.

Equilibria of the Rössler flow

For $\boldsymbol{y} = (y_1, y_2, y_3)$, the model used earlier is

$$\begin{aligned} y_1' &= -y_2 - y_3, \\ y_2' &= y_1 + a y_2, \\ y_3' &= b + y_3(y_1 - c), \end{aligned} \qquad a = b = 0.2, \quad c = 5.7.$$

- At equilibrium, $y_2^* = -y_3^*$ and $y_1^* = a y_3^*$.
- The last equation becomes $a(y_3^*)^2 - c y_3^* + b = 0$.
- Two equilibria from quadratic. The smaller y_3 is the center of the familiar spiral.

Lower Rössler equilibrium

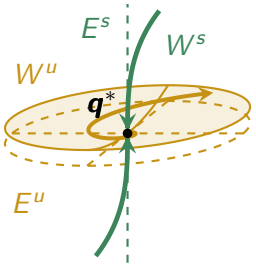
The Jacobian at $\boldsymbol{q}^* = (y_1^*, y_2^*, y_3^*)$ is

$$D\boldsymbol{f}(\boldsymbol{q}^*) = \begin{pmatrix} 0 & -1 & -1 \\ 1 & a & 0 \\ y_3^* & 0 & y_1^* - c \end{pmatrix}.$$

- For $(a, b, c) = (0.2, 0.2, 5.7)$, the lower equilibrium has one eigenvalue with negative real part and a complex pair with positive real part.
- W^s is locally a curve, while W^u is locally a surface.
- Along the stable direction, trajectories approach the equilibrium exponentially.
- The unstable directions spiral outward near the equilibrium.

Rössler manifold geometry

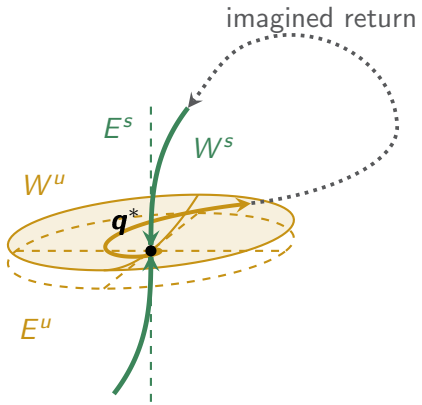
- On the stable curve, trajectories approach $\boldsymbol{q}^*$ as $t \to +\infty$.
- On the unstable surface, trajectories approach $\boldsymbol{q}^*$ as $t \to -\infty$.
- The Jacobian defines E^s and E^u, which approximate W^s and W^u near $\boldsymbol{q}^*$. Farther away, the manifolds bend.



Stable E^s, W^s; unstable E^u, W^u.

Can the spiral return?

- Trajectories on W^u spiral outward.
- W^u bends away from its tangent plane E^u.
- Could one trajectory loop back along W^s toward $\boldsymbol{q}^*$?



Dotted connector: hypothetical return.

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Online links: MIT poster guide; Minnesota poster guide; Manifold notes; Stability notes; Rössler (1976).